

Instructions: Use proper notation, and show all of your work to receive credit. Round numeric answers to at least 2 decimal places.

1. (1 pt.) True or False (circle your answer): If the p-value for your hypothesis test is larger than α , you should fail to reject H_0 .

True

False

Solution: True (Section 10.6 notes).

2. (4 pts.) Suppose you wish to conduct a large-sample hypothesis test at the 10% significance level for $\mu_1 - \mu_2$. The hypotheses are as follows: $H_0 : \mu_1 - \mu_2 = 0$ vs. $H_a : \mu_1 - \mu_2 > 0$. In terms of the test statistic, Z , what is the correct form of the rejection region? Circle your answer.

(a) $\{Z > z_{.05}\}$

(b) $\{Z > z_{.1}\}$

(c) $\{|Z| > z_{.05}\}$

(d) $\{|Z| > z_{.1}\}$

Solution: B (Section 10.3 notes).

3. (6 pts.) A real estate agent claims that less than 75% of retired couples prefer living in an apartment to owning their own home. Of 200 randomly sampled retired couples, 116 prefer living in an apartment. Test the agent's claim using $\alpha = 0.05$. Be sure to check the appropriate assumptions/conditions.

Solution: Section 10.3, Example 4.

Hypotheses: $H_0 : p = .75$ vs. $H_a : p < .75$

Assumptions:

1. We have a random sample, so the observations (couples) are independent.
2. $np_0 = 200(.75) = 150 \geq 10$ and $n(1 - p_0) = 200(1 - .75) = 50 \geq 10$

Test statistic: $Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{.58 - .75}{\sqrt{\frac{.75(.25)}{200}}} = -5.5522.$

You can use the rejection region OR p-value approach to make a decision:

Rejection region approach: Reject H_0 if $Z < -z_{.05}$; i.e., $Z < -1.645$. Since $Z = -5.5522 < -1.645$, we reject H_0 .

p-value approach: $P(Z \leq -5.5522) = \text{normalcdf}(-1E99, -5.5522, 0, 1) = 1.41 \times 10^{-8}$. Since $\text{p-value} \leq \alpha = .05$, we reject H_0 .

Conclusion: At the 5% significance level, there's enough evidence to conclude that the true proportion of retired couples who prefer living in an apartment to owning their own home is less than 0.75.

4. (8 pts.) An experimenter has prepared a drug dosage level that she claims will induce sleep for 80% of people suffering from insomnia. After examining the dosage, we feel that her claims regarding the effectiveness of the dosage are inflated. In an attempt to disprove her claim, we administer her prescribed dosage to 20 insomniacs and we observe Y , the number for whom the drug dose induces sleep. We wish to test the hypothesis $H_0 : p = .8$ versus the alternative, $H_a : p < .8$. Assume that the rejection region $\{y \leq 12\}$ is used.

(a) (4 pts.) Find α .

Solution: Section 10.1-10.2 HW, Problem 2b.

$$\begin{aligned}\alpha &= P(\text{Type I error}) \\ &= P(\text{Reject } H_0 \mid H_0 \text{ true}) \\ &= P(Y \leq 12 \mid p = .8) \\ &= \text{binomcdf}(20, .8, 12) \\ &\approx .0321\end{aligned}$$

(b) (4 pts.) Find β when $p = .6$.

Solution: Section 10.1-10.2 HW, Problem 2d.

$$\begin{aligned}\beta &= P(\text{Type II error}) \\ &= P(\text{Fail to reject } H_0 \mid H_0 \text{ false}) \\ &= P(Y > 12 \mid p = .6) \\ &= 1 - P(Y \leq 12 \mid p = .6) \\ &= 1 - \text{binomcdf}(20, .6, 12) \\ &\approx .4159\end{aligned}$$

5. (6 pts.) An organic chemist randomly sampled ten 4.85-g aniline specimens to produce acetanilide. The dry yield of acetanilide was measured for each specimen—the mean dry yield was 3.781 grams, and the standard deviation was 0.181 grams. Prior research shows that dry yields are normally distributed. The chemist wants to determine if there's evidence that the true mean dry yield of acetanilide differs from 4 grams. Test this claim using the p-value approach with $\alpha = 0.05$. Be sure to check the appropriate assumptions/conditions.

Solution: Section 10.8, Example 11.

Hypotheses: $H_0 : \mu = 4$ vs. $H_a : \mu \neq 4$

Assumptions:

1. Independent observations—we have a random sample of acetanilide specimens, so this is satisfied.
2. Dry yields are normally distributed.

Test statistic: $T = \frac{\bar{Y} - \mu_0}{s/\sqrt{n}} = \frac{3.781 - 4}{.181/\sqrt{10}} = -3.8262$.

p-value: $2 \cdot P(T \geq |-3.8262|) = 2 \cdot \text{tcdf}(3.8262, 1E99, 9) = .0041$.

Decision: Since $\text{p-value} = .0041 \leq \alpha = .05$, we reject H_0 .

Conclusion: At the 5% significance level, there's enough evidence to conclude that the true mean dry yield of acetanilide differs from 4 grams.